

Data Manipulation With Unix/Linux BASH

Note: *file* means *path/to/file* in the examples below

I want to ...		Example
Reshape data in single column (fasta, fastq)	Convert 1 column to 2 columns, row 1-2 become cols 1-2 Convert a FASTQ file to tidy, rows 1-4 become cols 1-4	<code>paste - - file</code> <code>paste - - - - file</code>
Wrangle Columns	Modify column delimiters (aka field separators) Keep Columns 1, 2, 3 and 6 in tsv file, remove the rest Keep Columns 1, 2, 3 and 6 in csv file, remove the rest Remove column names (first line) from any tidy file Bind columns together into same tidy file	<code>tr -s "PresentDelimiter" "NewDelimiter"</code> <code>cut -f1-3,6 file</code> <code>cut -d, -f1-3,6 file</code> <code>tail -n+2 file</code> <code>paste <(commands) <(commands)</code> <code>paste file1 file2</code>
Wrangle Rows	Return lines that match regex pattern Bind rows together from two sources into one tidy file Sort rows by column 2 alphabetically Sort rows by col3 from lrg to sml numbers Keep only consecutive unique values, optionally also count the number of duplicates Count the number of rows in a file, or the number of characters	<code>grep "Pattern" file</code> <code>cat <(commands) <(commands)</code> <code>cat file1 file2</code> <code>sort -k2,2 file</code> <code>sort -nrk3,3 file</code> <code>uniq file</code> <code>uniq -c file</code> <code>wc -l file</code> <code>wc -c file</code>
Modify values / text	Find & replace or remove text using regex pattern	<code>sed "s/FindPattern/ReplacePattern/g"</code>
Return multiple files using wildcards (globs)	Any and all characters repeated any number of times Any single character List all files ending with ".txt"	<code>*</code> <code>?</code> <code>ls *.txt</code>
Create regular expression search patterns	Escape character, change meaning of special character Match a special character like * or . 0-infinite repeats of the previous character Any single character Beginning of line or End of line Match any one of the following characters: a, b, c	<code>\</code> <code>* or \.</code> <code>*</code> <code>.</code> <code>^ or \$</code> <code>[abc]</code>
do math	add 1+1 and save to variable named X subtract 1-1 and save to variable name Y 1+11-2 and print to screen	<code>X=\$((1+1))</code> <code>Y=\$((10-5))</code> <code>echo \$((1+11-2))</code>

path to windows home dir from ubuntu: `/mnt/c/Users/YourWindowsUserName`